

SUMMARY

I am a Ph.D. student at UC Irvine studying Systems and Control under the supervision of Professor Faryar Jabbari. I have an education background in mechanical engineering, especially on optimization and control theory. My current interest is multi-agent systems, anti-windup control, event-triggered control, linear parameter varying systems, distributed optimization on graphs, and control barrier functions.

EDUCATION

Sep 2021 - now	Ph.D. candidate , Mechanical Engineering University of California, Irvine	3.941/4.0
Sep 2020 - Jun 2021	Master of Science , Mechanical Engineering University of California, Irvine	4.0/4.0
Sep 2016 - Jun 2020	Bachelor of Science , Automation Southeast University Exchange at UC Irvine in the fourth year	3.6/4.0

WORK EXPERIENCE

Mechanical Engineer UCI's Engineering Design in Industry Program	Arduino, Solidworks, 3D print	Jan 2020 - Mar 2020
- Lead development of innovative personal hygiene device for hot, damp, and sanitized towelette delivery - Collaborate with cross-functional team to optimize product functionality and user experience - More information on https://ethanyxu.com/WhoopyWipes		
Mechanical Engineer UCI's Engineering Design in Industry Program	Signal processing, Solidworks	Sep 2019 - Dec 2019
- Develop hands-free, noninvasive clinical solution for enhanced blood vessel visualization - Create comprehensive Bill of Materials (BOM) for efficient product manufacturing - Apply principles of objective and quantifiable assessment in medical device design		
Strategy consulting Intership PricewaterhouseCoopers	Strategy consult, Communication	Sep 2017 - Dec 2017
- Provide strategic consulting services to a prominent Electric and Electronic Manufacturing company - Utilize enterprise analysis techniques to identify market opportunities		

ACADEMIC EXPERIENCE

Graduate Student Lead Hacker Fab at UC Irvine	Mentoring, Photolithography, PCB	Sep 2024 - Now
- Lead interdisciplinary research team at UCI's branch of Hacker Fab (https://ethanyxu.com/IrvineHackerFab) - Spearhead development of optical lithography projects, e.g., tube furnace, spin coater, and patterning. - Drive fund-raising initiatives and oversee project management. Cultivate collaborative environment as graduate student leader (https://ethanyxu.com/UROP)		
Teaching Assistant / Instructor University of California, Irvine	Teaching, Communication	Sep 2021 - Now
- Assist Professor Faryar Jabbari in teaching ENGRMAE 80: Dynamics at University of California, Irvine - Deliver one-on-one support to enhance student understanding of complex dynamics concepts - Evaluate and grade homework assignments and exams, ensuring fair assessment		

AWARDS

CALIT2 IRT Graduate Student Mentor Stipend, University of California, Irvine	Mentoring, Project Management	Sep 2025 - May 2026
- Awarded a \$1,000 UCI stipend as graduate student mentor in the CALIT2 IRT Program - Coordinating project direction and cross-disciplinary collaboration		
Provincial Third Prize for National College Mathematical Contest in Modeling	Modelling, Optimisation	Sep 2018 - Dec 2018
- Lead team in designing regulation protocol for RGV robots in automated warehouse systems - Optimize working patterns to minimize robot stop time, enhancing overall efficiency		
Excellence Award for 20th electronic design contest of Southeast University	MCU development(STM32), Circuits	Apr 2018 - Jun 2018
- Collaborate in team to design feedback control system for DC power under varying load conditions - Test on bread board with oscilloscope, finalize with PCB board design		
First price for 14th RoboCup of Southeast University	ROS, Python, Pattern recognition	Dec 2017 - Jan 2018
- Program a robot using ROS for navigation and data collection - Use OpenCV in Python to recognize and track objects		

PUBLICATIONS AND PATENTS

[1] P. Ong, Y. Xu, R. M. Bena, F. Jabbari, and A. D. Ames, "Matrix Control Barrier Functions," *IEEE Transactions on Automatic Control*, pp. 1–13, Aug. 2026. doi: 10.1109/TAC.2026.3701302 Accessed: Jun. 10, 2026.

[2] Y. Xu and F. Jabbari, "Control of Heterogeneous Multi-Agent Systems with Active Leader," in *2026 American Control Conference (ACC)*, New Orleans, Aug. 2026. Accessed: Aug. 18, 2026.

[3] Y. Xu and F. Jabbari, "Distributed Optimization of Finite Condition Number for Laplacian Matrix in Multi-Agent Systems," *International Journal of Robust and Nonlinear Control*, vol. 36, no. 9, pp. 5030–5043, Jun. 2026. doi: 10.1002/rnc.70479

[4] Y. Xu and F. Jabbari, "Grid-LMIA: A Griding-Based Assembler for Solving Differentiable Parameter-Dependent Linear Matrix Inequalities," *arXiv preprint arXiv:2608.03175*, Aug. 2026. arXiv: 2608.03175. [Online]. Available: <https://arxiv.org/abs/2608.03175>

[5] Y. Xu and F. Jabbari, "Discrete-Time Leader-Following Multiagent Systems: Saturation Constraints and Event-Triggered Control," *IEEE Transactions on Control of Network Systems*, vol. 12, no. 2, pp. 1354–1368, Jun. 2025. doi: 10.1109/TCNS.2024.3516578

[6] Y. Xu and F. Jabbari, "Distributed Optimization of Network Weights for Improved Performance," in *2024 American Control Conference (ACC)*, IEEE, Jul. 2024, pp. 1392–1397. doi: 10.23919/ACC60939.2024.10644563

[7] Y. Xu and F. Jabbari, "Spline-based parameter varying output feedback synthesis with improved L_2 gain," in *2024 IEEE 63rd Conference on Decision and Control (CDC)*, IEEE, Dec. 2024, pp. 1455–1460. doi: 10.1109/CDC56724.2024.10886461

[8] Y. Xu and F. Jabbari, "Discrete-Time Output Feedback under Bounded Actuators: Single and Multi-agent Problems," in *2022 IEEE 61st Conference on Decision and Control (CDC)*, IEEE, Dec. 2022, pp. 4865–4871. doi: 10.1109/CDC51059.2022.9992896

[9] Y. Xu, Y. Chen, T. Liu, and W. Chen, "Optimization Algorithm for Power Flow Calculation Using Graph Theory," in *Lecture Notes in Electrical Engineering Proceedings of 2019 Chinese Intelligent Systems Conference*, 2020, pp. 142–150. doi: 10.1007/978-981-32-9698-5_17

[10] Y. Chen, W. Chen, Y. Xu, and T. Liu, "A New Method of Optimizing Newton-Raphson Power Flow Based on Graph decomposition," Oct. 2019. Accessed: Jul. 11, 2019.

PRESENTATION

- May 2026, "Control of Heterogeneous Multi-Agent Systems with Active Leader", Presented at 2026 American Control Conference (ACC), New Orleans, LA, USA Program
- April 2025, "Distributed optimization of the finite condition number of the Laplacian matrix", Presented at 45th Southern California Control Workshop, San Diego, CA, USA Program
- Dec 2024, "Spline-based parameter varying output feedback synthesis with improved L_2 gain", Presented at 2024 IEEE 63rd Conference on Decision and Control (CDC), Milano, Italy Presented Paper
- Jul 2024, "Distributed Optimization of Network Weights for Improved Performance", Presented at 2024 American Control Conference (ACC), Toronto, ON, Canada Presented Paper
- Dec 2022, "Discrete-Time Output Feedback under Bounded Actuators: Single and Multi-agent Problems", Presented at 2022 IEEE 61st Conference on Decision and Control (CDC), Cancun, Mexico Presented Paper
- Oct 2019, "Optimization Algorithm for Power Flow Calculation Using Graph Theory", Presented at 2019 Chinese Intelligent Systems Conference (CISC), Haikou, China Program

SKILLS

Program Language:	Matlab, Python, C/C++, Shell, LaTeX, HTML
Control Theory:	PID Control, Nonlinear Control, Robust Control (H_∞ , l_2 gain and gain scheduling), Event-Triggered Control, Control Barrier Function, Multi-Agent System, Anti-Windup Control, MPC, LQR
Mechanical Engineering:	3D printing, CAD(Solidworks), Lagrangian Mechanic, System Model via first principle, System Identification via Bode Plot, Linear Time-Invariant System,
Electric Engineering:	Micro-Controller Develop (Arduino, STM32, RaspberryPi), PCB desgin(Altium Designer, Easy EDA, KiCad), Oscilloscopes, Soldering, Circuits and Signals, Power and Energy System (AC,DC), DSP
Optimization:	Convex Optimization, Semi-definite Programming, Numerical Methods (Augmented Lagrangian, Interior Point Method, variations of Newton's method), ADMM

LANGUAGES

English:	Professional proficiency
Mandarin:	Native speaker